of thirty-eight, was looked upon as someone suspiciously long in the tooth to appear on the scene with radical new ideas like this.

Note the shift here from de Broglie's "matter waves" to Schrödinger's "wave function." Though Schrödinger was heavily influenced by de Broglie's work, his concept went quite a bit further, and deserves a distinct name. Most obviously, the value of a matter wave at any one point was some real number, while the amplitudes described by wave functions are complex numbers—the sum of a real number and an imaginary one.

More important, the original idea was that each kind of particle would be associated with a matter wave. That's not how Schrödinger's wave function works; you have just one function that depends on all the particles in the universe. It's that simple shift that leads to the world-altering phenomenon of quantum entanglement.

○ ○ ○

What made Schrödinger's ideas an instant hit was the equation he proposed, which governs how wave functions change with time. To a physicist, a good equation makes all the difference. It elevates a pretty-sounding idea ("particles have wave-like properties") to a rigorous, unforgiving framework. Unforgiving might sound like a bad quality in a person, but it's just what you want in a scientific theory. It's the feature that lets you make precise predictions. When we say that quantum textbooks spend a lot of time having students solve equations, it's mostly the Schrödinger equation we have in mind.

Schrödinger's equation is what a quantum version of Laplace's demon would be solving as it predicted the future of the universe. And while the original form in which Schrödinger wrote down his equation was meant for systems of individual particles, it's actually a very general idea that applies equally well to spins, fields, superstrings, or any other system you might want to describe using quantum mechanics.

Unlike matrix mechanics, which was expressed in terms of mathematical concepts most physicists at the time had never been exposed to, Schrödinger's wave equation was not all that different in form from Maxwell's electromagnetic equations that adorn T-shirts worn by physics students to this day. You could visualize a wave function, or at least you